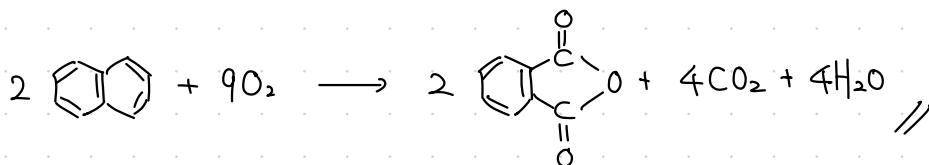
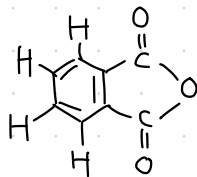
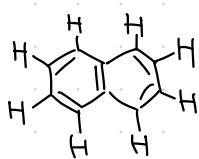
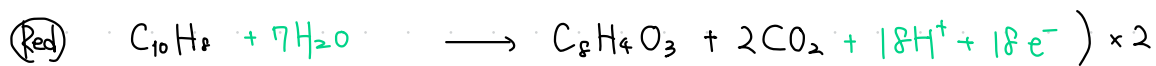
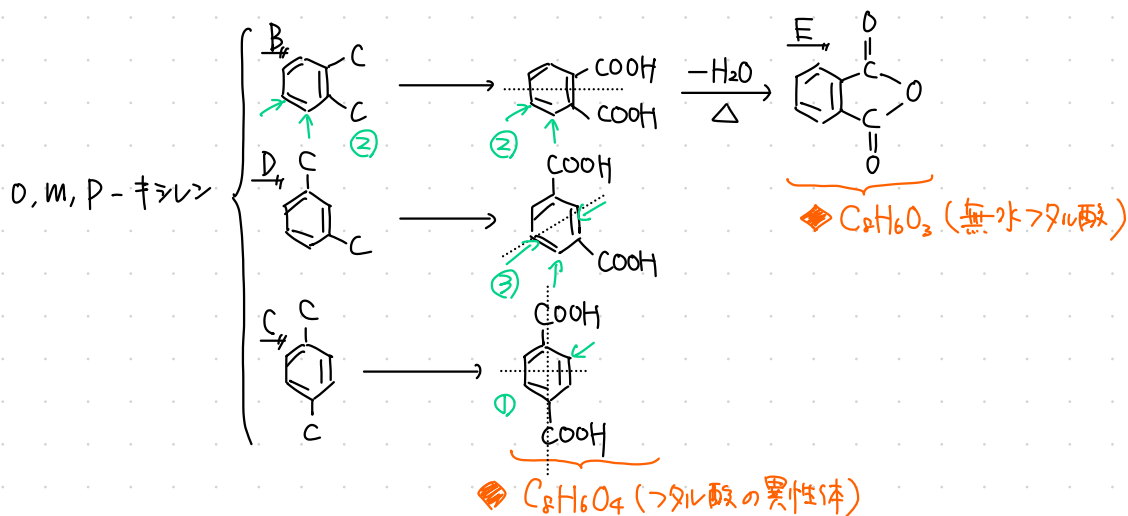
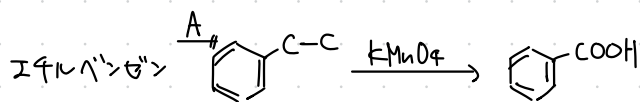


4-① C_8H_{10} $U = 4$ $\text{C}_6\text{H}_6 \times 1$

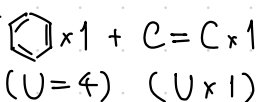


$\left(\begin{array}{l} \text{酸化だから, 余った } C \xrightarrow{O_2} CO_2, \\ H \xrightarrow{O_2} H_2O \text{ になると考え,} \\ \text{係数合わせでも良い} \end{array} \right)$

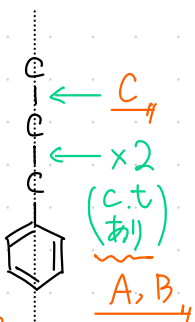
4-2

C_9H_{10}

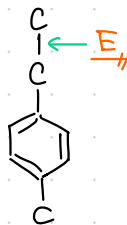
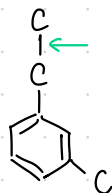
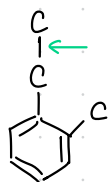
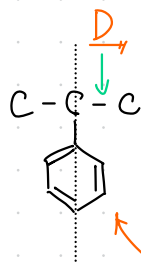
$U = 5$



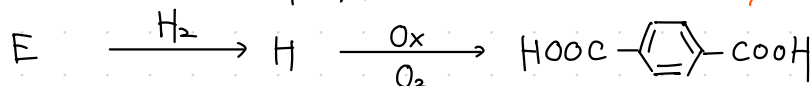
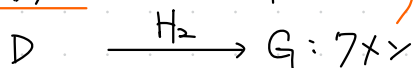
内2.



< 1置換体 >

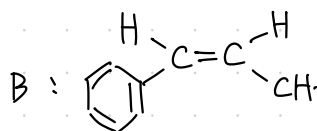
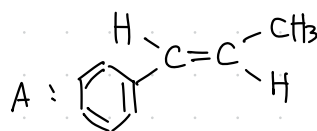


< 2置換体 >



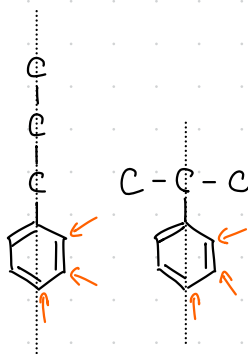
A, B: シス・トランス異性体

融点 $A > B$
 トランス シス



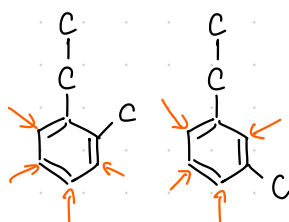
内1. 8

内3. 22



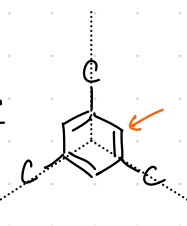
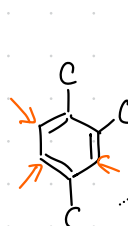
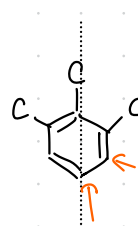
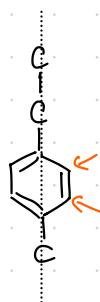
< 1置換体 >

Cl置換: 6



< 2置換体 >

Cl置換: 10



< 3置換体 >

Cl置換: 6

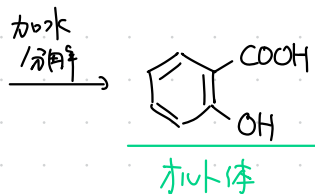
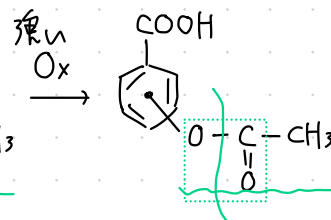
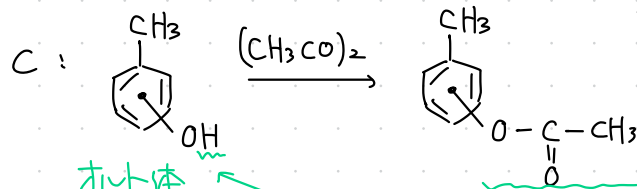
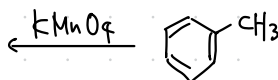
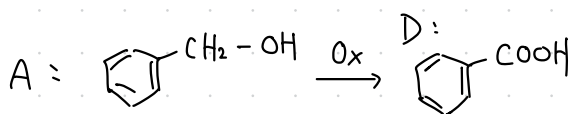
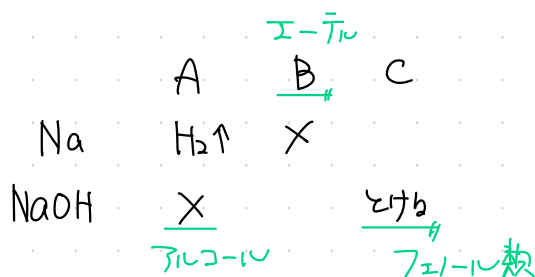
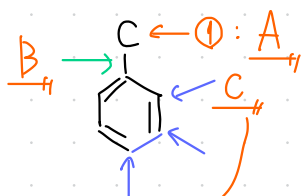
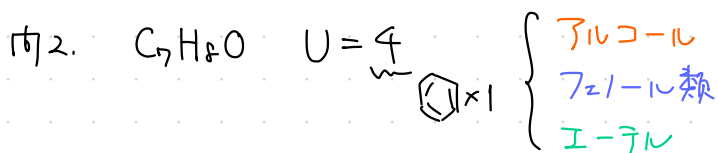
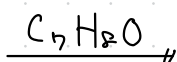
4-③



条件より $l : \frac{m}{2} = 7 : 4 \quad \therefore (l, m) = (7, 8)$

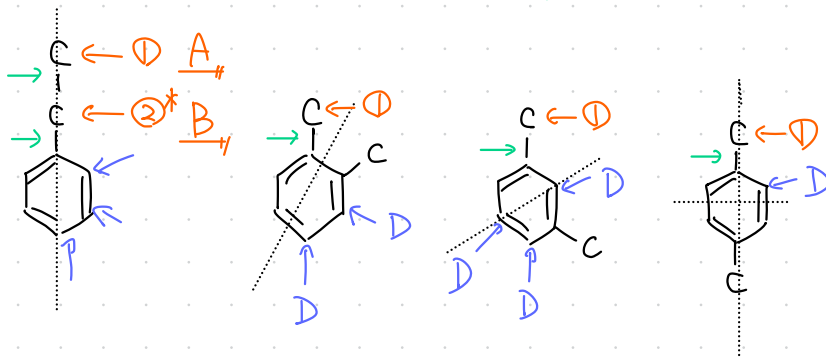
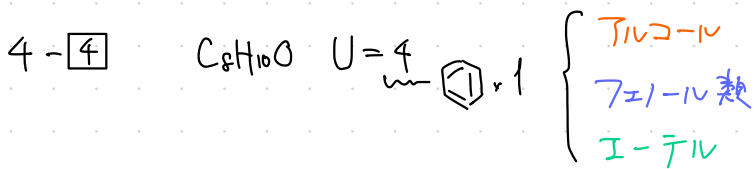
$\therefore (C_7H_8O_n)_k = (92 + 16n)k = 108$

$\therefore (n, k) = (1, 1)$

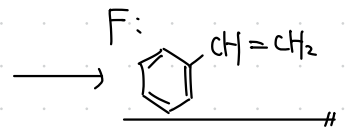
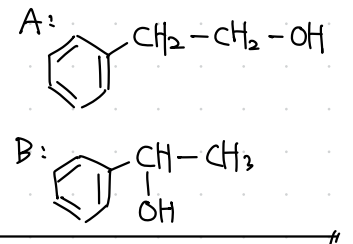
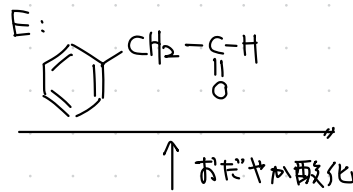
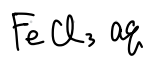
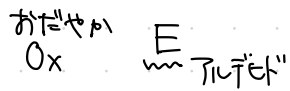
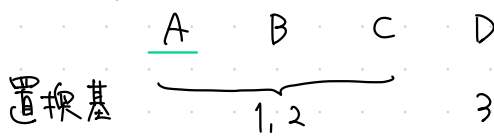


対体

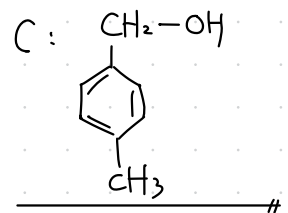
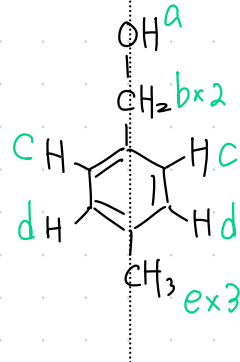
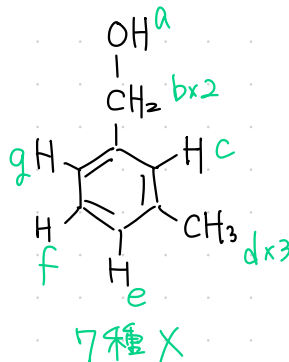
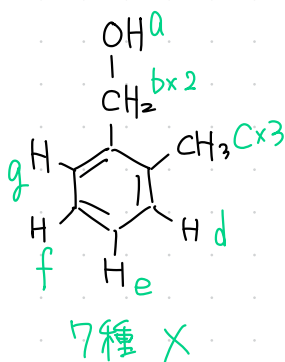
対体



第1級
アルコール



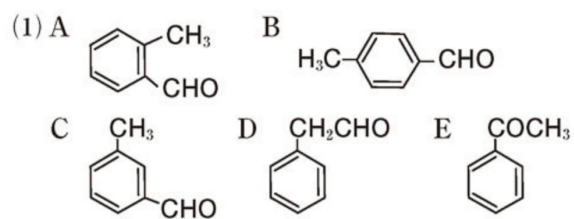
C: 1 or 2 置換体のアルコール



内2

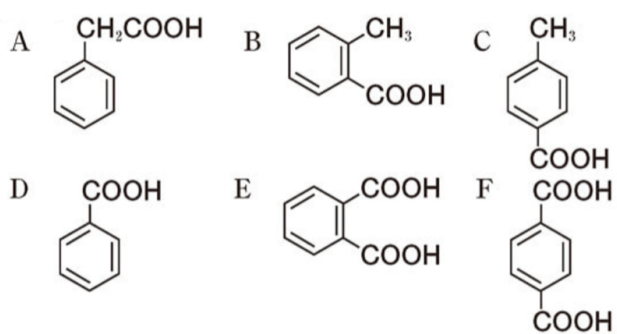
D: フェニル類 (3 置換体), $\underline{6}$

4 - 5

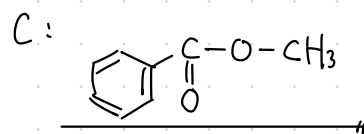
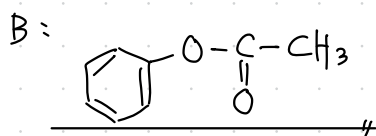
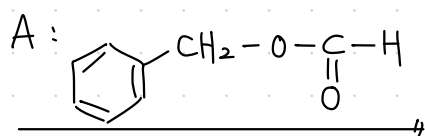
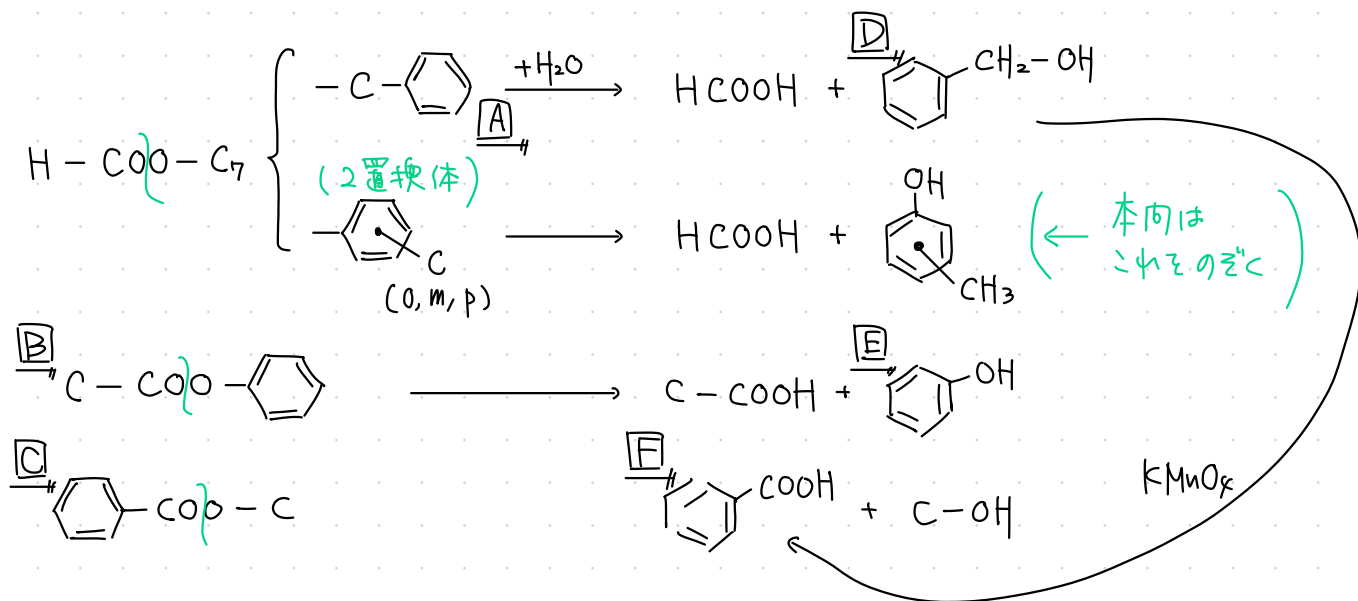
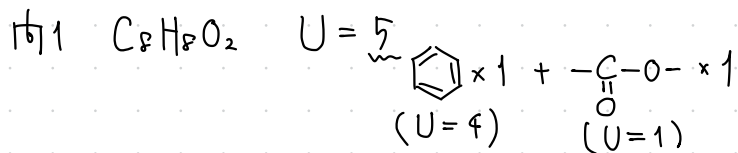


- (2) Gは*p*-置換体であるため、*o*-, *m*-置換体よりも分子の形が対称的である。したがって、結晶化しやすいため、融点は最も高くなる。

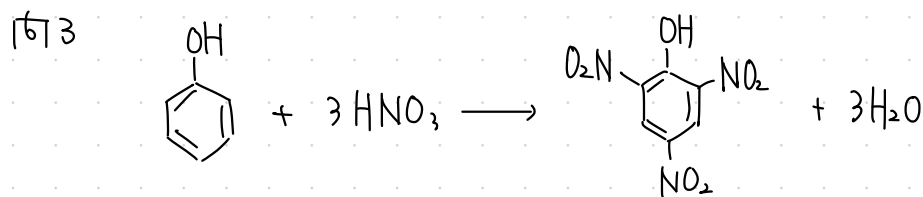
4 - 6



4-7



例2. E



トリニトロフェノール
 $(2, 4, 6 - \text{ト})$ ニトロフェノール

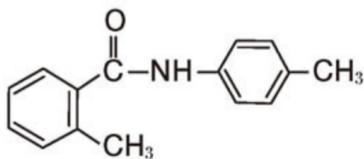
4-8

(1) $C_{15}H_{15}NO$

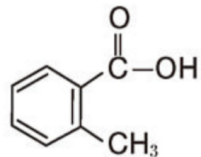
(2)

$$U = \frac{2 \times 15 + 2 + 1 - 15}{2} = 9$$

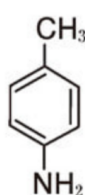
A



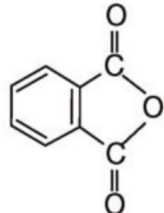
B



C



D



$C_{15}H_{15}NO$

$$U = \frac{2 \times 15 + 2 + 1 - 15}{2} = 9 = \text{アミド} - \text{C}(=\text{O})-\text{NH}- + \text{C}_6\text{H}_5 \times 2$$

アミド
A

加水分解
 H^+

①

カルボン酸
B
木片体

$KMnO_4$

B'

$-H_2O$
 Δ

酸無水物 D

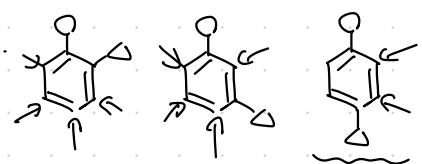
②

アミン C の HCl 塩

OH^-
WBP-リ

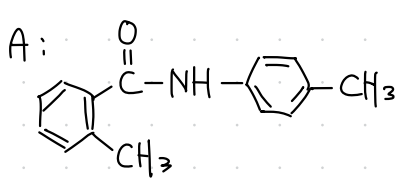
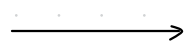
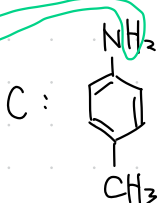
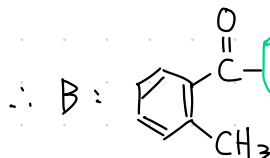
アミン C

2置換体

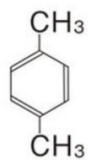


$-NH_2$,
 $-CH_3$ あり

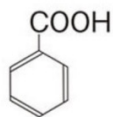
アミド A	カルボン酸 B	アミン C
C_{15}	C_8 以上	C_7 以上
	$\downarrow$ C_8	$\downarrow$ C_7



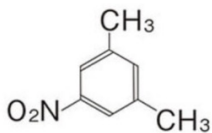
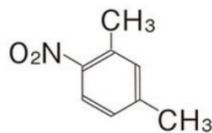
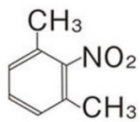
問 1 化合物 D :



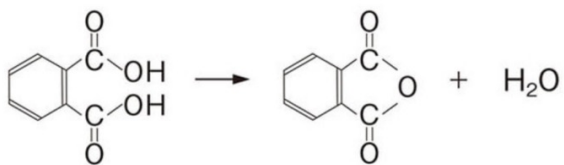
問 2 化合物 E :



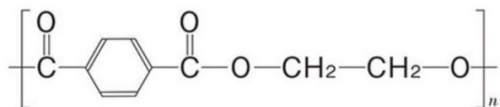
問 3



問 4



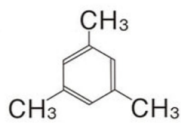
問 5



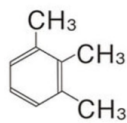
4 - 10

問1 (ア) エチレングリコール(1,2 エタンジオール) (イ) 酸素
(ウ) ヨウ素

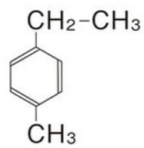
問2 A



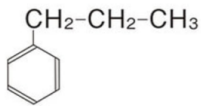
B



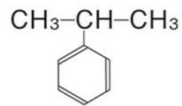
C



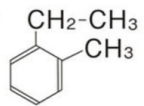
D



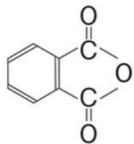
E



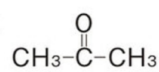
F



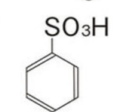
K



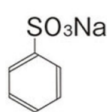
M



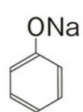
問3 N



O



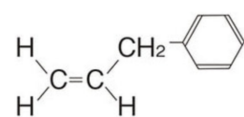
P



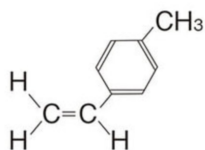
4 - 11

問1 2個 問2 クメン

問3A



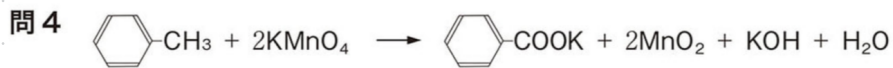
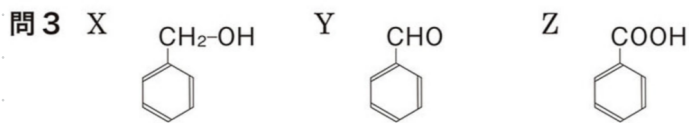
E



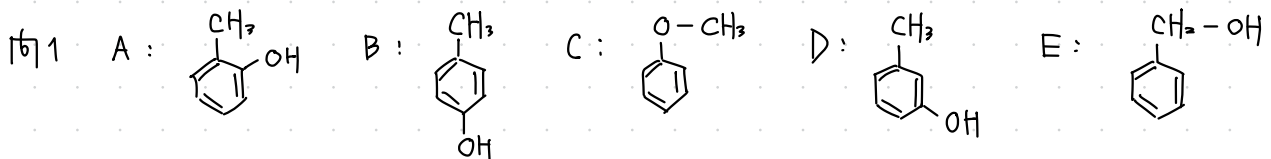
4-12

問1 気体…水素 官能基…ヒドロキシ基

問2 試薬… FeCl_3 色…紫



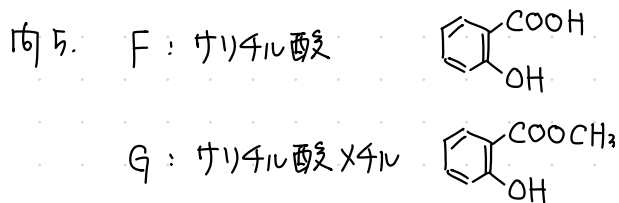
4-13



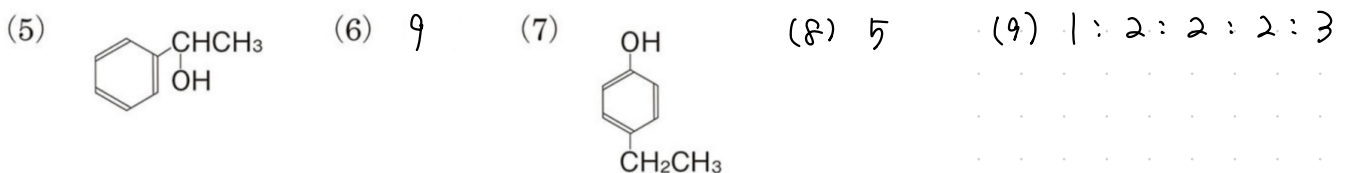
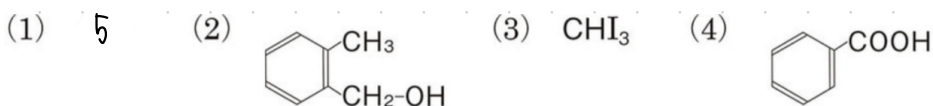
問2 NaOH aq を加えると A, B, D は 中和して均一な溶液になる (とける)。
(C, E は 2層に分離する) ↑ これは間違い。見ためや1においてわかることをかく。

問3 ヒドロキシ基を持つため、水素結合ができません, 分子間力が弱いから。

問4 金属 Na を加えたとき, C は H_2 を発生しない。

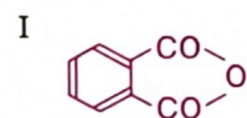
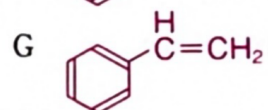
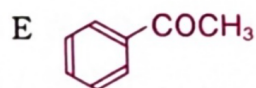
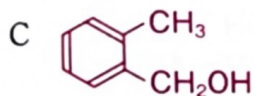
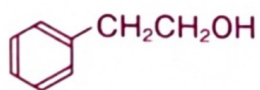


4-14

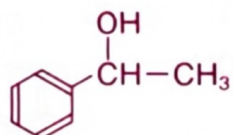


4 - 15

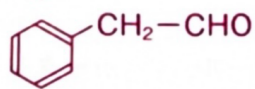
問1 A



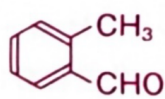
B



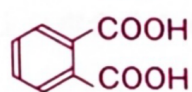
D



F



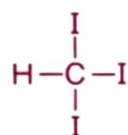
H



問2 B

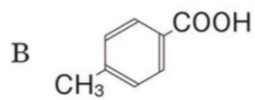
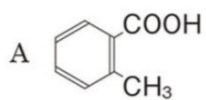
問3 ポリスチレン

問4 記号 B 構造式



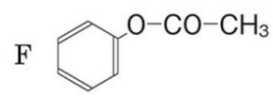
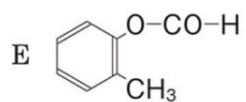
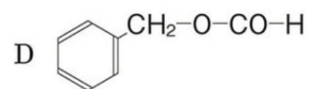
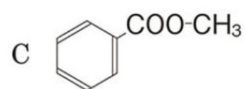
4 - 16

問1 (1)



(2) ナフタレン, *o*-キシレン

問2 (1)

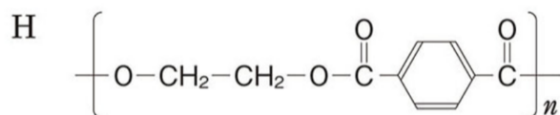
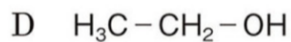
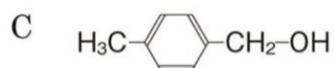
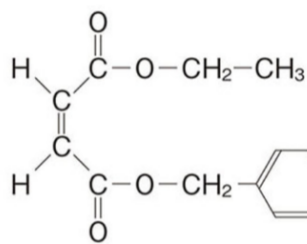


(2) 2,4,6-トリブロモフェノール

4 - 17

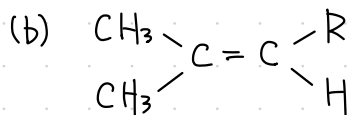
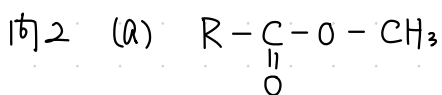
問1 $C_{14}H_{16}O_4$

問2 A

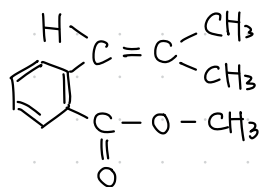


4 - 18

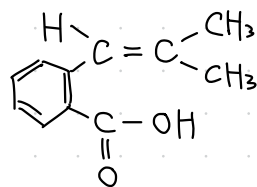
問1 $C_{12}H_{14}O_2$



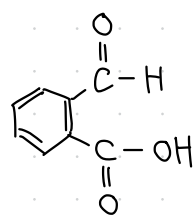
問3 A



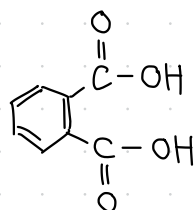
B



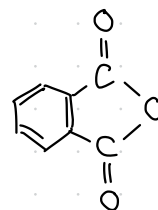
C



D

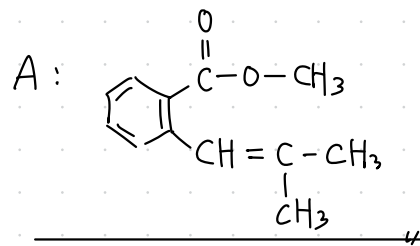
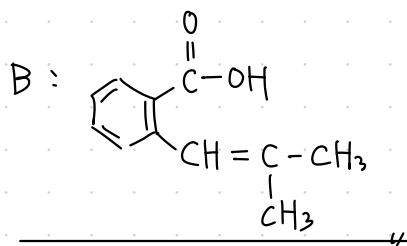
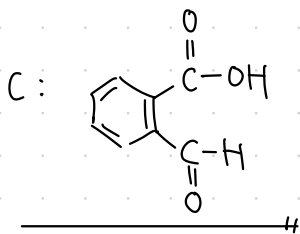
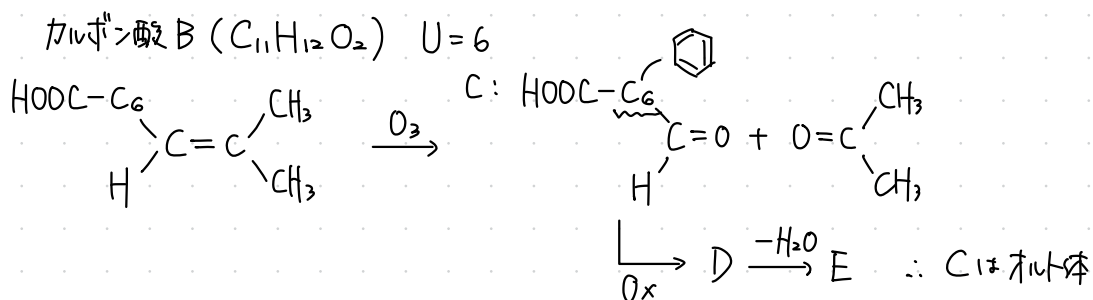
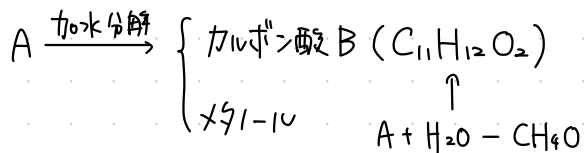
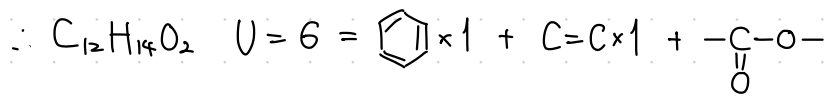


E



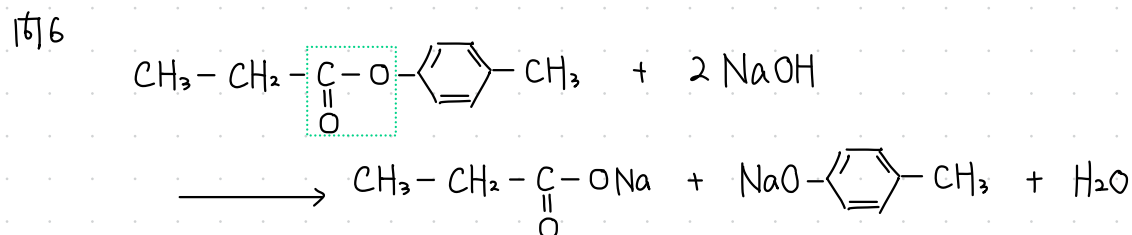
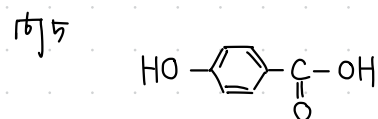
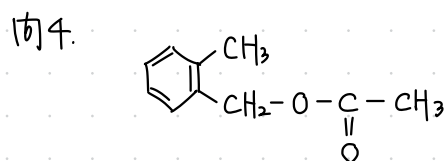
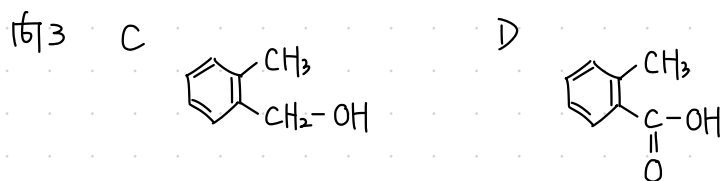
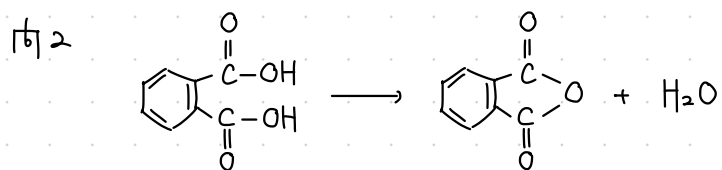
問1 $C_xH_yO_z$ について,

$$x = \frac{190 \text{ g} \times \frac{75.8}{100}}{12 \text{ g/mol}} = 12, \quad y = \frac{190 \text{ g} \times \frac{7.4}{100}}{1.0 \text{ g/mol}} = 14, \quad z = \frac{190 \text{ g} \times \frac{16.8}{100}}{16 \text{ g/mol}} = 2$$

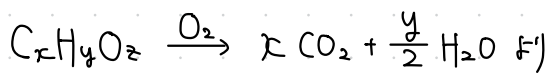


4 - 19

問1 あ C_5H_6O n 164 う $C_{10}H_{12}O_2$



問1 $\Delta f = k_f m$ $2.56 = 5.12 \times \frac{410 \times 10^{-3} \text{ g}}{M \text{ g/mol}} \times \frac{1}{5 \times 10^{-3} \text{ kg}}$ $\therefore M = 164 \text{ (u)}$



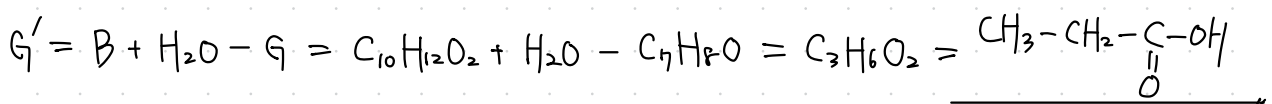
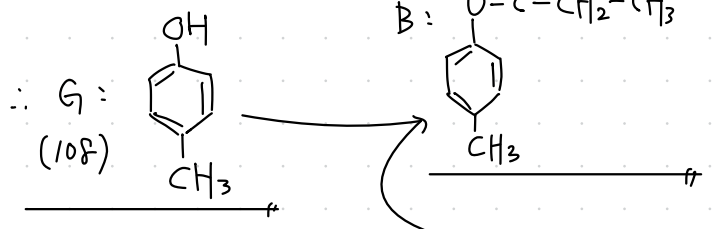
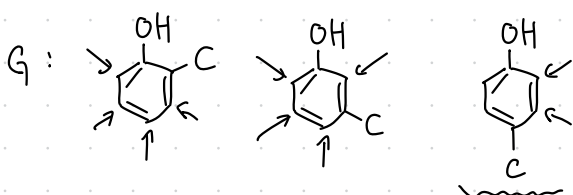
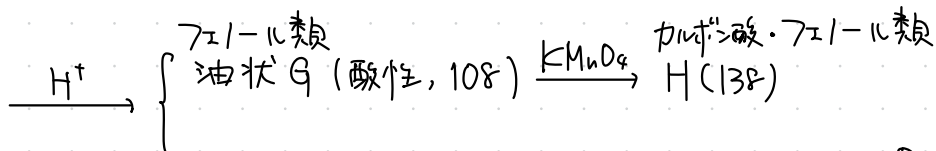
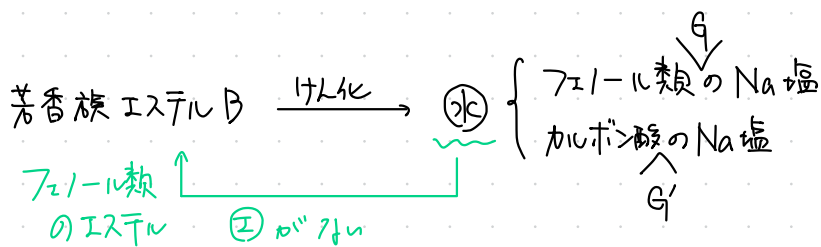
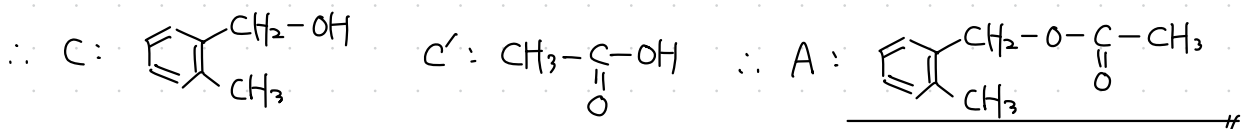
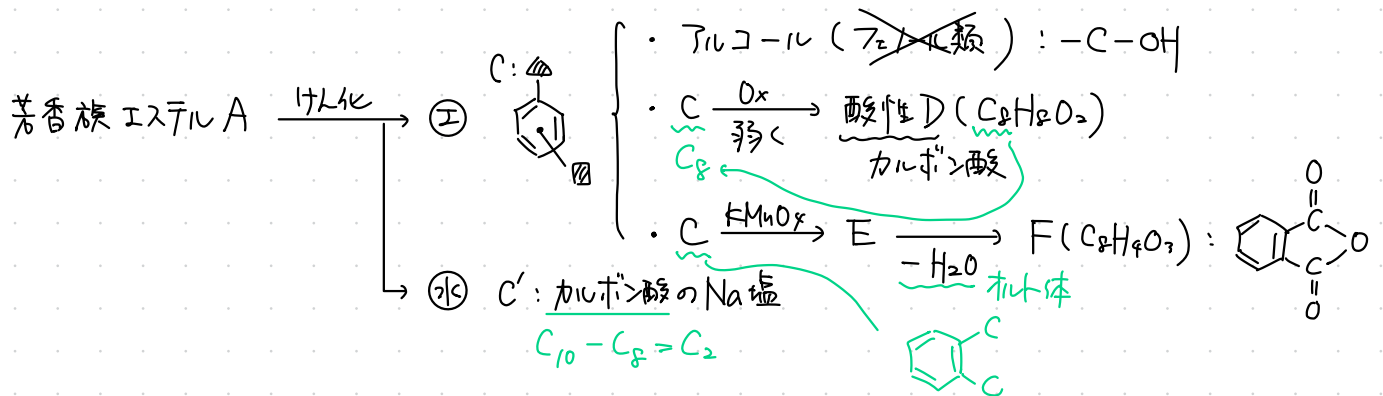
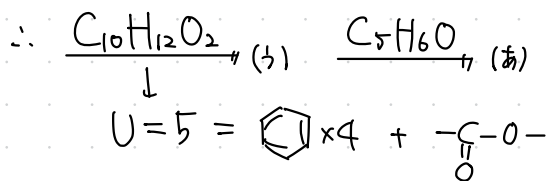
$C_x H_y O_z : CO_2 = 1 : x$ したがって,

$\frac{8.2 \text{ g}}{164 \text{ g/mol}} \times x = \frac{22 \text{ g}}{44 \text{ g/mol}} \therefore x = 10$

$C_x H_y O_z : H_2O = 1 : \frac{y}{2}$ したがって,

$\frac{8.2 \text{ g}}{164 \text{ g/mol}} \times \frac{y}{2} = \frac{5.4 \text{ g}}{18 \text{ g/mol}} \therefore y = 12$

$M = 12x + y + 16z = 120 + 12 + 16z = 164 \therefore z = 2$

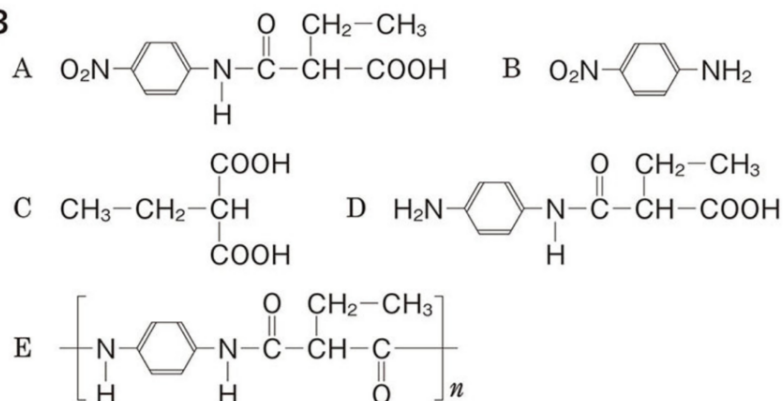


4- 20

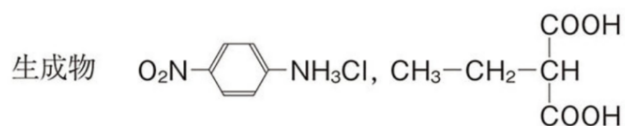
問1 C 原子…52.4 % H 原子…4.8 % N 原子…11.1 %

問2 $\text{RCOOH} + \text{NaHCO}_3 \rightarrow \text{RCOONa} + \text{CO}_2 + \text{H}_2\text{O}$

問3

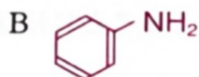
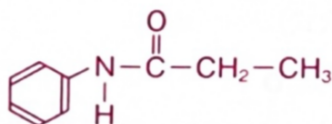


問4 塩酸を用いることで化合物 A のアミド結合が加水分解されてしまうため。

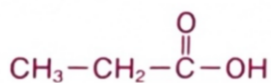


4- 21

問1 A

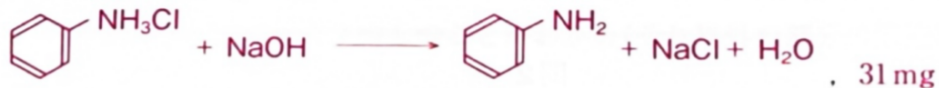


C



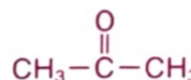
D $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-OH}$

問2



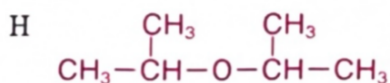
問3

F $\text{CH}_3\text{-CH}_2\text{-O-CH}_3$, エチルメチルエーテル

G , アセトン

問4 F, E, D

問5



I $\text{CH}_2=\text{CH-CH}_3$

問1 $C_{19}H_{20}O_4$

